

FNCE 5352 — Lecture 5 (Part 1): Classification & Logistic Regression

Regression vs Classification

- Definition:
 - **Regression:** Predicts continuous outcomes. It estimates a mapping function (f) from input variables (X) to a continuous output variable (Y).
 - **Classification:** Predicts discrete outcomes. It categorizes input data into two or more classes.
- Output Type:
 - **Regression:** The output is a real or continuous value (e.g., salary, price).
 - **Classification:** The output is a category (e.g., default or no default).
- Both regression and classification are types of supervised machine learning algorithms, where a model is trained according to the existing model along with correctly labeled data

RMSE

- Definition: The square root of the average of the squared differences between the predicted and actual values. It measures the standard deviation of the residuals (prediction errors).
- Interpretation: Represents the average error made by the model in predicting the outcome. Lower values indicate better fit.
- Scale: Depends on the target variable's scale. Higher RMSE values may not necessarily indicate a poor model, especially if the target variable has a wide range.
- Sensitivity: More sensitive to outliers than other metrics, such as MAE (Mean Absolute Error).

RMSE Formula

- $\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$
- n is the number of observations.
- y_i is the actual value of the observation.
- $\hat{y}_i$ is the predicted value.
- The sum $\sum_{i=1}^n (y_i - \hat{y}_i)^2$ calculates the squared differences between the actual and predicted values.

R^2

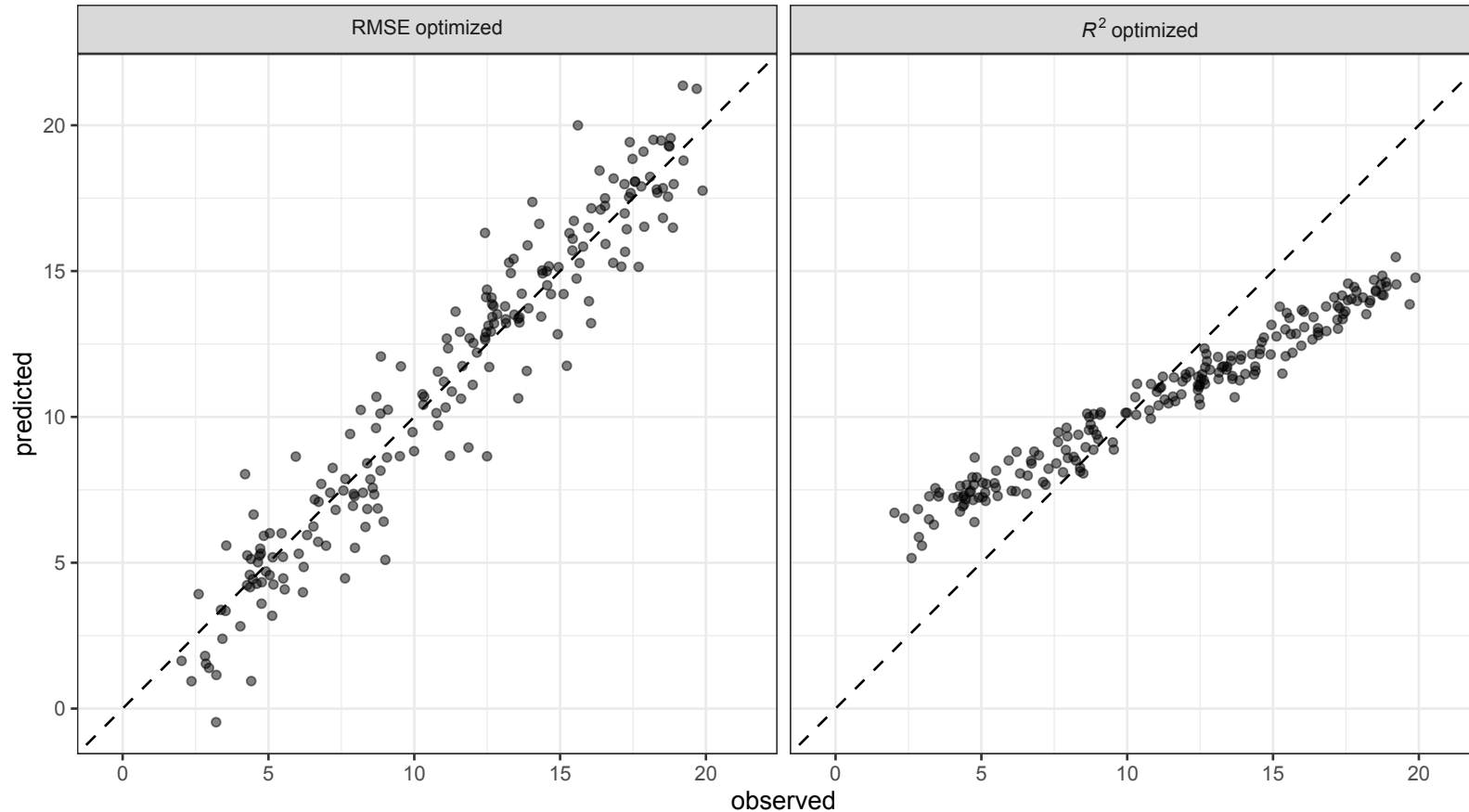
- Definition: The proportion of the variance in the dependent variable that is predictable from the independent variables. It is a statistical measure of how close the data are to the fitted regression line.
- Interpretation: Values range from 0 to 1. A higher value indicates a better fit, with 1 meaning the model explains all the variability of the response data around its mean.
- Scale-Independent: Unlike RMSE, R-squared is a normalized measure, making it easier to compare the goodness of fit across different datasets and models.
- Limitation: Can be misleadingly high in models with many predictors or when using higher-order polynomials. Adjusted R-squared is often used to account for the number of predictors.

R^2 Formula

- $R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$
- n , y_i and $\hat{y}_i$ are defined the same as in *RMSE*
- The numerator, $\sum_{i=1}^n (y_i - \hat{y}_i)^2$, is the sum of the squared differences between the actual and the predicted values (also known as the sum of squares of residuals).
- The denominator, $\sum_{i=1}^n (y_i - \bar{y})^2$, is the total sum of squares (TSS) or the sum of the squared differences between the actual values and the mean of the actual values.

RMSE vs R^2

RMSE measures *accuracy* while R^2 measures *correlation*.



Observed versus predicted values for models that are optimized using the RMSE compared to the coefficient of determination

Calculating RMSE using yardstick

```
1 library (yardstick)
2
3 rmse(data_with_predictions, truth = latency, estimate = .pred)
```

```
#> # A tibble: 1 × 3
#>   .metric .estimator .estimate
#>   <chr>   <chr>      <dbl>
#> 1 rmse    standard      59.2
```

Calculating Multiple Metrics

```
1 metrics <- metric_set(rmse, rsq, mae)
2 metrics(data_with_predictions, truth = latency, estimate = .pred)
```

```
#> # A tibble: 3 × 3
#>   .metric .estimator .estimate
#>   <chr>    <chr>      <dbl>
#> 1 rmse    standard    59.2
#> 2 rsq     standard     0.380
#> 3 mae     standard    40.2
```

- RMSE: difference between the predicted and observed values 
- R^2 : squared correlation between the predicted and observed values 
- MAE: similar to RMSE, but mean absolute error 

Classification

- Qualitative variables take values in an unordered set $\mathcal{C}$, such as:
 $\text{eye color} \in \{\text{brown}, \text{blue}, \text{green}\}$
 $\text{email} \in \{\text{spam}, \text{ham}\}.$
- Given a feature vector X and a qualitative response Y taking values in the set $\mathcal{C}$, the classification task is to build a function $C(X)$ that takes as input the feature vector X and predicts its value for Y ; i.e. $C(X) \in \mathcal{C}$.
- Often we are more interested in estimating the *probabilities* that X belongs to each category in $\mathcal{C}$.

For example, it is more valuable to have an estimate of the probability that an insurance claim is fraudulent, than a classification fraudulent or not.

Classification: hard vs soft predictions

- **Soft prediction:** estimate a probability
 - $\hat{p}(X) = P(Y = 1 \mid X)$
- **Hard prediction:** turn that probability into a class label
 - $\hat{y}(X) = \mathbf{1}\{\hat{p}(X) > t\}$ for some threshold t

Why soft predictions matter in credit risk

- We can **rank** accounts by risk (scorecards)
- We can choose different actions at different probability ranges
 - e.g., approve / approve-with-conditions / manual review / decline
- Thresholds change when costs or objectives change

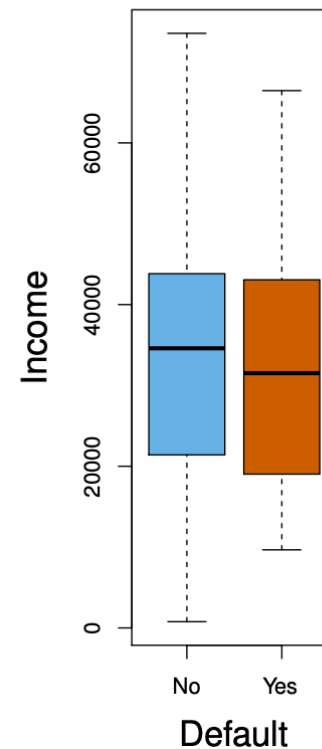
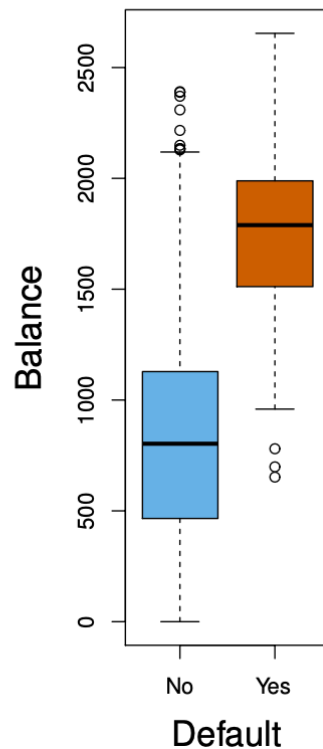
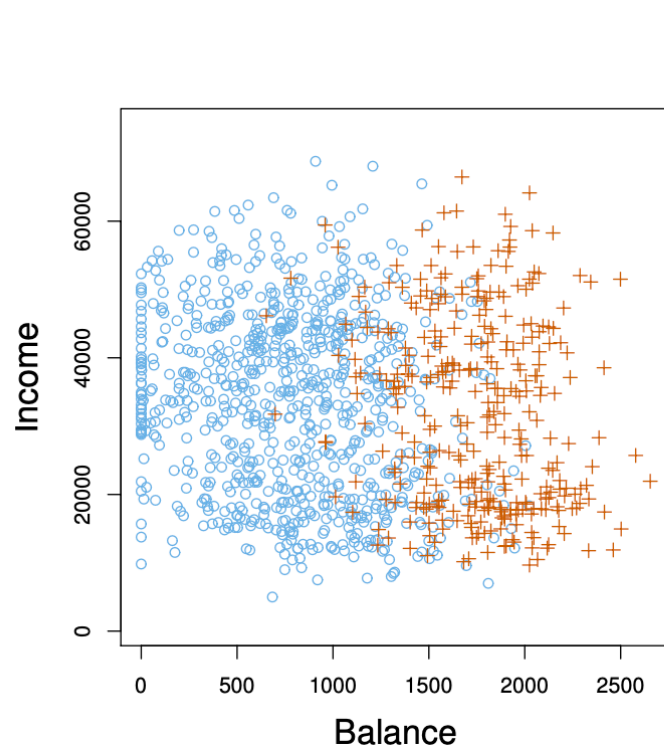
Thresholds are business decisions

- Picking $t = 0.50$ is **not** automatically “right”
- You pick t based on tradeoffs:
 - false positive vs false negative costs
 - capacity constraints (manual reviews)
 - regulatory / policy constraints

Example: credit card / default-style framing

- $Y = 1$: "bad outcome" (fraud / default / delinquency / downgrade)
- X : features available at decision time (balance, utilization, income, behavior, etc.)
- Output: $\hat{p} = P(Y = 1 \mid X)$

Example: Credit Card Default



Can we use Linear Regression?

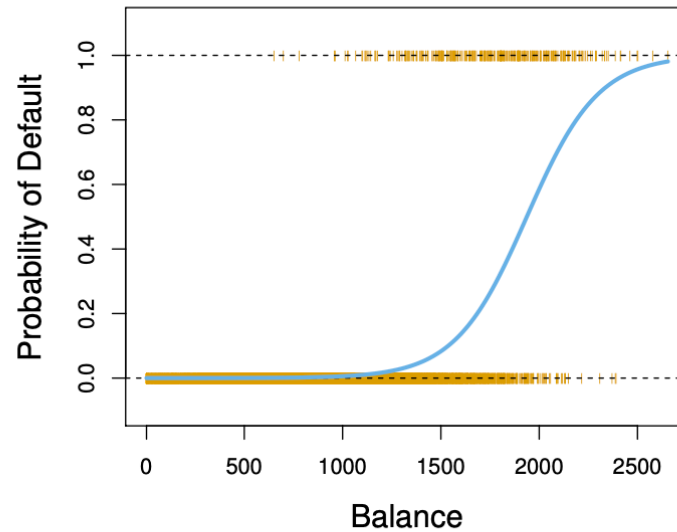
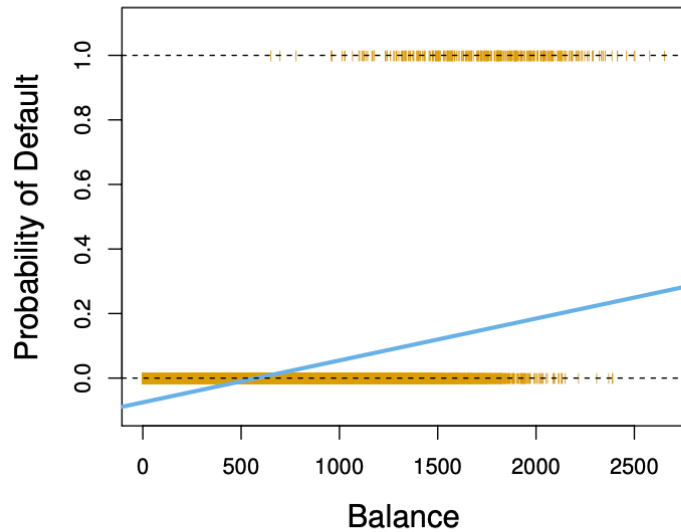
Suppose for the **Default** classification task that we code

$$Y = \begin{cases} 0 & \text{if No} \\ 1 & \text{if Yes.} \end{cases}$$

Can we simply perform a linear regression of Y on X and classify as **Yes** if $\hat{Y} > 0.5$?

- In this case of a binary outcome, linear regression does a good job as a classifier, and is equivalent to *linear discriminant analysis* which we discuss later.
- Since in the population $E(Y|X = x) = \Pr(Y = 1|X = x)$, we might think that regression is perfect for this task.
- However, *linear* regression might produce probabilities less than zero or bigger than one. *Logistic regression* is more appropriate.

Linear versus Logistic Regression



The orange marks indicate the response Y , either 0 or 1. Linear regression does not estimate $\Pr(Y = 1|X)$ well. Logistic regression seems well suited to the task.

Why not just use linear regression?

- Linear regression can predict values < 0 or > 1
- The relationship between probability and predictors is rarely linear
- Classification wants a model constrained to $[0, 1]$

Linear vs logistic (big idea)

- Logistic regression models probability using an S-shaped curve
- Think: "linear model on log-odds"

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Fitting logistic regression in R

Small snippet only — we'll do the real work in the [.R](#) demo file.

```
1 fit <- glm(default ~ balance + income + student,  
2           data = train,  
3           family = binomial)  
4  
5 p_hat <- predict(fit, newdata = test, type = "response")
```

Interpreting coefficients

- Sign: increases vs decreases risk
- Magnitude: relates to odds, not raw probability
- Optional: odds ratio for a 1-unit increase in X_j is $\exp(\beta_j)$

Prediction = probability, not a decision

- Model output: $\hat{p}_i = P(Y_i = 1 \mid X_i)$
- Decision rule needs a threshold t :
 - predict 1 if $\hat{p}_i > t$

Part 1 wrap

You should be able to:

- explain why logistic regression is used for 0/1 outcomes
- fit a simple logistic regression with `glm()`
- generate predicted probabilities with `predict(..., type="response")`

Next (after break): how to evaluate + choose thresholds.